

Aejandro De La Torre

LIS 440

Lecture: wk12: Lecture-Demo - More Subsetting and Selection in Pandas

11/18/25

Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path

# Print version of pandas just for tracking purposes
print(pd.__version__)

# Because the professor does this
print("Alejandro De La Torre's Code...")

# Set current working directory (CWD) : root folder
root = Path(__file__).parent.parent.resolve()

# Show CWD
print("Current Working Directory: ", Path.cwd())

# Set CWD: Data
data = root / 'data'
# Set CWD: Data > Raw
data_raw = data / 'raw'
# Set CWD: Data > Processed
data_processed = root / 'data' / 'processed'
# Set CWD: Scripts
scripts = root / 'scripts'

# Read Randomized Ice Cream Data
icecream_data = data_processed / 'icecreamWeek_sp25_AlejandroDeLaTorre.csv'

# Create dataframe of Ice Cream (IC) data
ic_df = pd.read_csv(icecream_data)
print(ic_df)

# Define labels for rows
rowLabels = ["choc", "straw", "van", "bmoon", "orng", "chry", "cdough"]

# View first and last 3 entries for vibes
print("First 3 rows: \n", ic_df.head(3))
print("Last 3 rows: \n", ic_df.tail(3))

print()
print("New row labels:")
print(rowLabels)
ic_df.index = rowLabels
print()
print("Reindexed data frame:")
```

```

print(ic_df)

flavorSeries = ic_df["Flavors"]
print()
print ("Flavor series:")
print (flavorSeries)

sortedFlavorSeries = flavorSeries.sort_values()
print()
print("Sorted Flavor Series:")
print(sortedFlavorSeries)

print()
sugarConeSeries = ic_df["Sugar Cone"]
print("sugarConeSeries:")
print(sugarConeSeries)

print()
chocolateRow = ic_df.iloc[0]
print("Chocolate Row:")
print(chocolateRow)

print()
VanillaRow = ic_df.loc["van"]
print("Vanilla Row:")
print(VanillaRow)

print()
VanillaRow = ic_df.iloc[2]
print("Vanilla Row:")
print(VanillaRow)

print()
cupSeries = ic_df["Cup"]
print("Cup Sales")
print(cupSeries)
print("Cup Strawberry Sales:")
cupStrawberrySales = cupSeries.iloc[1]
print(cupStrawberrySales)
cupStrawberrySales = cupSeries.loc["straw"]
print(cupStrawberrySales)

cones_df = ic_df[ ["Sugar Cone", "Waffle Cone"] ]
print()
print("Cones Sales, Sugar and Waffle:")
print(cones_df)

cones_col = ["Sugar Cone", "Waffle Cone"]
print()
print("Cone columns:")
print(cones_col)
cones_dfic_df = ic_df[ cones_col ]
print("Cone Sales:")
print(cones_df)

```

Shell

```
>>> %Run lec_icecream_sortcol.py
```

2.3.3

Alejandro De La Torre's Code...

Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 12/Tuesday/lec_activity/scripts

```
    Flavors Sugar Cone Cup Waffle Cone
```

```
0  Chocolate    40 31    23
```

```
1  Strawberry    25 42    27
```

```
2   Vanilla    40 37    42
```

```
3  Blue Moon    20 30    25
```

```
4   Orange    25 30    11
```

```
5   Cherry    20 22    43
```

```
6 Cookie Dough    40 10    25
```

First 3 rows:

```
    Flavors Sugar Cone Cup Waffle Cone
```

```
0  Chocolate    40 31    23
```

```
1  Strawberry    25 42    27
```

```
2   Vanilla    40 37    42
```

Last 3 rows:

```
    Flavors Sugar Cone Cup Waffle Cone
```

```
4   Orange    25 30    11
```

```
5   Cherry    20 22    43
```

```
6 Cookie Dough    40 10    25
```

New row labels:

```
['choc', 'straw', 'van', 'bmoon', 'orng', 'chry', 'cdough']
```

Reindexed data frame:

```
    Flavors Sugar Cone Cup Waffle Cone
```

```
choc  Chocolate    40 31    23
```

```
straw  Strawberry    25 42    27
```

```
van    Vanilla    40 37    42
```

```
bmoon  Blue Moon    20 30    25
```

```
orng   Orange    25 30    11
```

```
chry   Cherry    20 22    43
```

```
cdough Cookie Dough    40 10    25
```

Flavor series:

```
choc    Chocolate
```

```
straw   Strawberry
```

```
van     Vanilla
```

```
bmoon   Blue Moon
```

```
orng    Orange
```

```
chry    Cherry
```

```
cdough  Cookie Dough
```

Name: Flavors, dtype: object

Sorted Flavor Series:

```
bmoon   Blue Moon
```

```
chry    Cherry
```

```
choc    Chocolate
```

```
cdough  Cookie Dough
```

orng Orange
straw Strawberry
van Vanilla
Name: Flavors, dtype: object

sugarConeSeries:
choc 40
straw 25
van 40
bmoon 20
orng 25
chry 20
cdough 40
Name: Sugar Cone, dtype: int64

Chocolate Row:
Flavors Chocolate
Sugar Cone 40
Cup 31
Waffle Cone 23
Name: choc, dtype: object

Vanilla Row:
Flavors Vanilla
Sugar Cone 40
Cup 37
Waffle Cone 42
Name: van, dtype: object

Vanilla Row:
Flavors Vanilla
Sugar Cone 40
Cup 37
Waffle Cone 42
Name: van, dtype: object

Cup Sales
choc 31
straw 42
van 37
bmoon 30
orng 30
chry 22
cdough 10
Name: Cup, dtype: int64
Cup Strawberry Sales:
42
42

Cones Sales, Sugar and Waffle:
 Sugar Cone Waffle Cone
choc 40 23
straw 25 27

van	40	42
bmoon	20	25
orng	25	11
chry	20	43
cdough	40	25

Cone columns:

['Sugar Cone', 'Waffle Cone']

Cone Sales:

	Sugar Cone	Waffle Cone
choc	40	23
straw	25	27
van	40	42
bmoon	20	25
orng	25	11
chry	20	43
cdough	40	25

>>> >>>